

Algorithmic Impact Assessment Results

Version: 1.0.1

Section 1: Impact Level : 2

Current Score: 45

Raw Impact Score: 45

Mitigation Score: 58

Section 2: Requirements Specific to Impact Level 2 Notice

Plain language notice posted through all service delivery channels in use (Internet, in person, mail or telephone).

Explanation

The explanation from level I is published.

In addition, a more detailed, meaningful, plain language, explanation is provided to the client with any decision that results in the denial of a benefit or service, or involves a regulatory action.

This explanation must inform the client of the reason or justification of the administrative decision. This involves a clear and client-focused description of how the automated decision system came to the output it did, including:

- the principal factors that led to it, such as, where appropriate, a description of the decision tree, scoring or weights of certain factors, and
- how the automated decision system output was used by human officers

The client must also be provided with the link to the published level I explanation.

Peer review

Consult at least one of the following qualified experts and publish the complete review or a plain language summary on a Government of Canada website:

- Experts from a federal, provincial, territorial or municipal government institution
- Faculty members of a post-secondary institution
- Researchers from a relevant non-governmental organization
- Contracted third-party vendor with a relevant specialization
- A data and automation advisory board specified by Treasury Board of Canada Secretariat

Gender-based Analysis Plus

Ensure that the [Gender-based Analysis Plus](#) has been consulted with your department's diversity and inclusion experts and includes:

- An assessment of how the automation project might impact different population groups. This

includes consideration of the impacts of the automated decision system and data used in the project, as well as the likely impact of the final decision. Where possible, cite the data used to assess the impacts. It is recommended that the data be gender-disaggregated and include other intersecting identity factors such as age, disability and race. If the data is unavailable, identify where the data gaps exist

- Details of planned or existing measures to address risks identified through the Gender-based Analysis Plus or other assessments

Training

Role-based training on how to appropriately use and explain the functionalities of the automated decision system, at a high level.

Ensuring human involvement

The system may make decisions and assessments without direct human involvement.

Humans are to be involved in system quality assurance and can intervene in the making of decisions by the system where appropriate.

Approval for the system to operate

Assistant deputy minister responsible for the program.

Other requirements

The Directive on Automated Decision-Making also includes other requirements that must be met for all impact levels.

[Link to the Directive on Automated Decision-Making](#)

Contact your institution's ATIP office to discuss the requirement for a Privacy Impact Assessment as per the Directive on Privacy Impact Assessment.

Section 3: Questions and Answers

Section 3.1: Project Details

Project Details

1. Name of Respondent

Elisabeth Mantil

2. Job Title

A/Science Leader

3. Name of ADM responsible for the program using the automated decision system

Debbie Beresford-Green

4. Department

Canadian Food Inspection Agency

5. Branch

Food Safety and Science Branch

6. Teams consulted on completion of the Algorithmic Impact Assessment
Data Engineering Services, Risk Management Outcomes and Modernization

7. Project Title
Establishment-based Risk Assessment Model for Food Establishments (ERA-Food Model)

8. Project ID from IT Plan
1617-020

9. Service ID
N/A

10. Departmental Program (from Departmental Results Framework)
Setting Rules for Food Safety and Consumer Protection

11. Project Phase
Implementation

[Points: 0]

12. Provide a project description:
The Canadian Food Inspection Agency (CFIA) is committed to leveraging its data, intelligence reports and surveillance information more effectively to identify emerging trends. This approach enables the agency to prioritize risks and strengthen program design, planning, compliance, and enforcement activities. Under the Safe Food for Canadians Regulations (SFCR) and within the scope of the modernization of its inspection system, the CFIA has developed a quantitative risk assessment model, known as the Establishment-based Risk Assessment Model for domestic food establishments (ERA-Food model) to better allocate, manage, and prioritize its inspection resources.

The ERA-Food model considers common food safety hazards and helps determine the appropriate level of oversight needed to manage them. It uses a mathematical algorithm to evaluate the risk associated with products manufactured by federally regulated domestic food establishments and their potential impact on Canadian consumers. By combining scientific data, establishment-specific information, and compliance history collected by CFIA inspectors, the model assesses the potential human health impact associated with each establishment.

The mathematical algorithm considers sixteen (16) risk factors identified through a literature review and validated by scientific experts. These fall into three categories: inherent factors related to the food commodity, operation, or manufacturing process; mitigation factors that represent strategies to reduce specific hazards; and compliance factors based on inspection results and the establishment's compliance with regulatory requirements. The ERA Food model generates risk results that can be further analyzed into value added insights. It generates a food safety risk profile by summarizing the factors influencing risk. Profiles are created both for individual establishments and for broader sectors (e.g., dairy).

Section 3.2: Impact Questions and Answers

Reasons for Automation

1. What is motivating your team to introduce automation into this decision-making process?

(Check all that apply)

Improve overall quality of decisions

Perform tasks that humans could not accomplish in a reasonable period of time

Use innovative approaches

Other (specify)

2. Please describe

Ensure consistency on the risk assessment process by following a science-based approach.

3. What are the client needs the system will address? Describe how client needs have been identified.

The system addresses the CFIA's tactical work planning team's (i.e., the clients) needs for risk assessment of federally regulated domestic food establishments. The ERA Automated Business solution has been implemented since April 2021, resulting into an agency-wide system whose outputs are accessible to a wide range of internal users (e.g. risk managers, food inspectors, risk intelligence analysts). The automated solution includes processes and functions to adequately manage the input of information and risk results. Data for the ERA model is automatically extracted from internal systems according to established business rules and controls to ensure consistency and data integrity. Internal stakeholders' needs were identified through various communication channels (e.g. information sessions, trainings, email inquiries). Outputs are tailored to these needs, informing clients on the food safety risk associated to establishments. The goal is for these tools to inform the CFIA's decision making processes.

4. Describe any positive impacts on clients and Canadian society that the system is expected to have.

Benefits associated with having a risk-based automated system for CFIA clients include:

- Enhancing the Agency's effectiveness with respect to program design and resource allocation;
- Strengthening preventive and evidence-based decision-making to food safety oversight in Canada that is flexible according to emerging and scientific trends and to changes within the Canadian food industry, ultimately increasing the public's confidence in this system;

Benefits for external clients (i.e. Safe Food for Canadians (SFC) licence holders operating food establishments with domestic activities) include:

- Increasing transparency and fairness to regulated parties by sharing individual and collective information to identify areas for improvement in compliance and mitigation approaches; and
- Bringing the Canadian system in line with the inspection approaches used by Canada's trading partners.

In addition, the following strategic benefit affects a much broader stakeholder group, including the Government of Canada and Canadians:

- CFIA can demonstrate to Canadians and trading partners that the oversight system for food is effective through ongoing rigorous and systematic assessment of risk and the use of meaningful performance indicators.

5. How effective will the system likely be in meeting client needs?

Very effective

[Points: +0]

6. Describe why effectiveness is as selected.

The ERA-Food model helps the CFIA prioritize its oversight activities and allocate its inspection resources using a scientific, risk-based decision-making approach. By evaluating food establishments based on the level of risk their products represent to Canadian consumers, the model provides input to the agency's risk management process. This information is taken into consideration when developing the program direction and determining the prioritization and frequency of inspection activities and work planning. The ERA-Food model is therefore a consistent and transparent assessment process for regulated parties, ensuring establishments with similar levels of risk are treated with similar rigor.

7. Describe how you will ensure that the system is confined to addressing the client needs identified.

The model is designed to remain strictly aligned with its intended purpose of supporting CFIA's risk-based oversight mandate and addressing clients needs related to inspection prioritization and resource allocation.

The ERA-Food model supports internal clients (e.g. CFIA inspectors, analysts, risk managers) by providing a scientific, consistent, and risk-based assessment of food establishments. Through controlled access with IBM Cognos Analytics, internal users can generate risk profiles that inform program direction, inspection frequency, and work planning. Defined access controls and governance procedures ensure that the model is used only for these defined operational and decision-making purposes.

The model has also been designed to meet the needs of external clients (i.e. regulated establishments) by producing transparent reports (i.e., risk profiles) that clearly explain what information was considered in the risk calculation. With a standardized risk assessment methodology, establishments presenting similar levels of risk are treated with similar rigor. CFIA inspectors then share copies of the profiles with regulated parties during site visits. This ensures the information is accurate, clearly communicated, and limited to supporting compliance and awareness, without enabling unintended or inappropriate use.

By constraining model functionality to risk assessment and oversight planning, and implementing controls around access, interpretation, and communication of outputs, the ERA-Food model ensures it remains focused on its defined clients needs. Ongoing monitoring, governance, and feedback mechanisms further ensure that the system continues to support both internal operational effectiveness and external transparency.

8. Describe any trade-offs between client interests and program objectives that you have considered during the design of the project.

A central design element of this project was having a standardized questionnaire to collect information directly from regulated parties. For external clients (i.e. regulated parties), this activity might be perceived as burdensome and not always well received. However, for internal clients, this information is essential to ensure consistent, high-quality data inputs for the ERA-Food model, to enable accurate risk assessment.

As of October 2025, regulated parties with domestic activities must complete the Food Safety Supplemental Information (FSSI, previously named Additional Establishment Information (AEI)) to obtain, renew or amend an SFC license.

To mitigate the administrative trade-off associated with this requirement, the questionnaire was designed using plain language and structured to collect only necessary information. The questionnaire's purpose, how the data will be used, and the benefits for transparency and consistency in oversight were clearly communicated. This approach balances program objectives with client impact by ensuring the integrity and effectiveness of the ERA-Food model while taking steps to minimize burden and enhance transparency for regulated parties. The approach has confirmed buy-in from industry as demonstrated by high completion rates (over 80% and increasing).

9. Describe any improvements, benefits, or advantages your department expects from using an automated system.

- Data not already collected by existing CFIA applications/systems are centralized into one system.
- The ERA Risk Profile is available both through an internal reporting interface and will be available through a web service that external users would be able to access.
- Capacity to extract aggregated data for internal and external usage.
- Adaptable to the other CFIA's business lines (i.e. Animal and Plant health).
- Data is available in near real-time.

10. What are your current program indicators, service standards or performance targets?

Performance of the ERA-Food model is currently assessed via a Treasury Board (TB) indicator (code IFS-R2-1) under the "Setting Rules for Food Safety and Consumer Protection" program. This indicator measures the extent to which regulated establishments are integrated into the ERA-Food model, serving as a proxy for data coverage and model uptake (i.e., through the completion of the FSSI questionnaire) by external stakeholders. Target values for each commodity sector depend on the number of years the commodity has been integrated into the model.

Additional enhancements to the CFIA food program allow measuring how inspection activities are aligned with risk-based prioritization over time. Since the implementation of this indicator, it is now possible to monitor the extent to which inspections are informed by risk-based insights derived from the ERA-Food model's outputs.

Together, these indicators support monitoring model coverage, and the effective use of risk information in inspection planning, contributing to the accuracy, consistency, and transparency of risk-based oversight.

11. What is your current performance level against these indicators, standards or targets?

Current performance level against the ERA Treasury Board indicator (IFS-R2-1) showed that three sectors met or exceeded their targets (i.e., fresh fruits and vegetables, dairy, and meat). The remaining six sectors (manufactured food, fish, processed fruits and vegetables, egg, maple, and honey) did not meet their targets, although positive performance trends were observed in all cases. Notably, the fish, honey, egg, and maple sector were within 5% of their targets, indicating near achievement.

Performance levels are influenced by factors such as the rate at which regulated parties provide the required information, as well as sector-specific characteristics (e.g. size, maturity, and seasonality). Overall, current performance of this indicator demonstrates improving data coverage and

increasing completion rates of the FSSI questionnaire by external clients, which supports the continued strengthening of the ERA-Food model. Further progress is expected as regulatory requirements and stakeholder participation evolves.

For the performance metrics on how well inspection activities are delivered according to risk, current results show a strong alignment between risk levels and inspection effort, particularly for establishments categorized as higher risk. For example, the proportion of inspection cases delivered by CFIA inspectors increased from 0% to 100% for risk category 1 establishments, from 72% to 93% for category 2 establishments, and from 77% to 84% for category 3 establishments. These results demonstrate improved targeting of inspection resources toward higher-risk establishments and increased consistency in the application of risk-based oversight.

12. How do you expect the system will change performance against your program indicators, service standards or performance targets?

Because as of October 2025, regulated parties with domestic activities must complete the Food Safety Supplemental Information (FSSI) to get a licence, this change is expected to significantly improve data completeness for the ERA-Food model (and consequently, positively impact the IFS-R2-1 indicator performance). This means, the model will be able to assess 100% of domestic food establishments within the next two years, strengthening the accuracy, consistency, and coverage of risk assessments.

As near-complete data coverage is achieved, the IFS-R2-1 indicator will become less relevant. Performance measurement is expected to evolve towards indicators that focus on the effective use of risk intelligence (e.g. alignment of inspection activities with risk levels) and the overall impact of the model on risk-based oversight. In this context, continued progress is expected in the consistent application of risk-based oversight, with inspection activities increasingly aligned to higher-risk establishments, supporting more targeted and effective use of inspection resources.

13. Have alternative non-automated processes been considered?

Yes

[Points: +0]

14. If non-automated processes were considered, why was automation identified as the preferred option?

If a non-automated process is used to run the ERA-Food model, several drawbacks will be expected including:

- The algorithm would be applied to the data using custom developed scripts and not in an automated system.
- Risk results would not be refreshed in near real-time but on an annual basis.
- Public reporting would be delayed.
- No ability to verify automatically that the information centralized is complete.
- The information provided by regulated parties would require more human intervention for data curation which reduces data quality and timeliness.

15. What would be the consequence of not deploying the system?

Service cannot be delivered in a timely or efficient manner

Service quality is not as high

Service delivery cannot achieve performance targets

Risk Profile

16. Is the project within an area of intense public scrutiny (for example, because of privacy concerns) and/or frequent litigation?

No [Points: +0]

17. Does the line of business serve equity denied groups?

No [Points: +0]

18. Are there any population groups that the automated decision system should not be applied to?

No [Points: +0]

19. Has the system been assessed to understand whether it would create or exacerbate barriers for persons with disabilities, and are measures in place to address any identified issues?

Assessment has been conducted, barriers were identified, and measures are in place or planned to remove or prevent them. [Points: +0]

20. If barriers for persons with disabilities were identified, describe the barriers and any planned or existing measures to identify, remove, and/or prevent them.

N/A - Barriers for persons with disabilities were not identified.

21. Could the automated decision-system be at heightened risk of being an exploitation target for malicious actors (for example, fraud for financial gain)?

No [Points: +0]

22. Have potential issues or harms with the existing service delivery approach been raised by clients or their representatives, the media, audits or parliamentary committees?

No [Points: +0]

23. Have you submitted a concept case to TBS and presented the project to the Government of Canada Enterprise Architecture Review Board?

No [Points: +0]

Project Authority

24. Will you require new policy or legal authority for this project?

No [Points: +0]

25. Indicate why you have chosen the response.

The ERA-Food model operates under the Safe Food for Canadians Act and Regulations (SFCR) as a tool to support implementation of SFCR inspection requirements. It refines oversight under the existing licensing framework—particularly the periodic inspections already mandated under SFCR.

Data feeding into ERA Food model comes through processes already authorized under SFCR and agency policy:

- The Food Safety Supplemental Information (FSSI, formerly the Additional Establishment Information (AEI) questionnaire) is submitted via MyCFIA as part of existing licensing/renewal processes.
- Compliance history is accessed via CFIA's databases.

About The System

26. Check which of the following capabilities apply to your system.

Risk assessment: Analyzing large data sets to identify patterns and recommend courses of action and in some cases trigger specific actions
Process optimization and workflow automation: Analyzing large data sets to identify any anomalies, cluster patterns, predict outcomes or ways to optimize; and automate specific workflows

27. Who developed the system?

Your institution [Points: +0]

28. Who configured the system?

Your institution [Points: +0]

29. Is the system:

Custom built [Points: +1]

30. Have you assigned accountability in your institution for the design, development, and maintenance, of the system?

Yes [Points: +0]

31. Is there a group in your institution responsible for identifying and implementing opportunities for improvement?

Yes [Points: +0]

32. Does your system conform with the appropriate clauses of standard CAN/ASC EN 301 549:2024, Accessibility requirements for Information and Communication Technology (ICT) products and services?

Yes [Points: +0]

33. Is a non-automated alternative planned and available in the event the automated decision system was unavailable for an extended period of time?

Yes [Points: +0]

34. Is there a process in place to grant, monitor, and revoke access permission to the system?

Yes [Points: +0]

35. The impacts that the development, operation, and maintenance of the system could have on the ongoing sustainability of an environmental ecosystem, may be:

Little to no impact [Points: +1]

36. Describe why the impacts resulting from the system's development, operations and maintenance are as selected.

The impacts resulting from the system's development, operations, and maintenance have little to no effect on the ongoing sustainability of an environmental ecosystem. The ERA-Food model only requires the energy and resources associated with standard information technology infrastructure and IT systems.

About the Algorithm

37. The algorithm used will be a (trade) secret

No [Points: +0]

38. The algorithmic process will be difficult to interpret or to explain

Yes [Points: +3]

39. Is the algorithm (Check all that apply):

Based on rules developed from laws and policy and/or using human expertise
(for example, if/then statements)

40. If the automated decision-system is model based, is the model inference (select all that apply):

Deterministic (follow precise rules and generate single outcome)

41. Is the model:

Discriminative (for example, regression analyses, decision trees)

[Points: +0]

42. Describe the model being used.

The ERA-Food model uses a deterministic and multiplicative structure. The model generates a single, fixed risk estimate for each establishment, rather than a range or uncertainty interval. Criterion weights linked to relevant risk factors of a given establishment are multiplied within each cluster (inherent risk, mitigation and compliance) and sub-cluster to produce multiple combined weights, which are then adjusted to the maximum risk increase/reduction for their corresponding sub-cluster. These adjusted weights are lastly multiplied by the establishment's initial risk, to produce the model's point estimate. A logarithmic scale is then used to establish the seven risk categories. The higher the result, the greater the food safety risk to Canadian consumers.

43. Explain why you chose the model

This model was chosen and developed by CFIA in collaboration with a Scientific Advisory Committee (SAC) composed of members from academia (University of Montreal, University of Manitoba, Dalhousie University, and University of Guelph), as well as representatives from Health Canada, the Public Health Agency of Canada, and a consultant from Risk Sciences International. The model was designed based on information gained through an international benchmarking exercise that evaluated risk assessment tools developed by various countries to assess the risk of food establishments. This included insights on lessons learned by the countries throughout the implementation of their tools and their application to inform resource allocation.

44. Will the algorithm continue to learn and evolve as it is used?

No [Points: +0]

45. Does the algorithm consider protected characteristics to make its decisions or recommendations?

No [Points: +0]

46. Have you evaluated whether variables that the system bases its decisions or recommendations on could be proxies for protected characteristics?

No [Points: +2]

47. Describe why you have not evaluated whether variables could be proxies for protected characteristics.

An evaluation of whether variables in the model could act as proxies for protected characteristics was not conducted because the model does not rely on any demographic, socio-economic, or personal identifiers that could reasonably correlate with protected characteristics such as race, gender, age, or ethnicity. Since none of the input variables capture information about individuals or groups in a way that relates to these attributes, the likelihood that any variable could function as a proxy is minimal. In the absence of data that could be linked – directly or indirectly to protected characteristics, there is no meaningful basis for conducting a proxy-bias assessment.

About the Decision

48. Describe the administrative decision(s) that is being supported or made by the algorithm.

Five (5) main functions of the ERA technical solution:

1. Data Collection: populates the ERA system with the required risk factors' information as well as other contextual information by extracting data from existing systems.
2. Algorithm calculation: computes the risk result for an establishment by using the risk factors populated by the data collection function.
3. Risk results: stores the calculated risk result number and retain the value for historical purposes and trends analysis.
4. Reporting management: generates outputs and reports at the individual and aggregated levels which includes the risk result and profile and related information to support decision making and provide input to stakeholders' business processes.
5. Data governance: enables the configuration of the data used to support the model.

49. Does the decision pertain to any of the categories below (check all that apply). Note that the examples in brackets are not exhaustive.

Health-related services (environmental and workplace health, product and food safety, drug regulation)

[Points: +1]

Impact Assessment

50. Which of the following best describes the type of automation you are planning?

Partial automation (the system will contribute to administrative decision-making by supporting an officer with assessments, recommendations, intermediate decisions, or other outputs)

[Points: +2]

51. Describe the role of the system in the decision-making process.

The ERA technical solution assesses the food safety risk associated with regulated establishments with domestic activities. Risk managers use the model outputs along with additional risk intelligence to guide CFIA's program design and allocation of inspection resources.

52. Will the system be making decisions or assessments that require judgment or discretion?

No

[Points: +0]

53. Describe.

The ERA-Food model is designed to be objective, standardized and

reproducible in the quantification of food safety risk associated to federally regulated food establishments based on data and predefined criteria. It uses measurable inputs and applies an established algorithm with weighted risk factors to produce a consistent risk assessment. Outputs of this model are one of many factors taken into consideration during the development of tactical work planning at the agency. Indeed, CFIA inspectors retain sole authority for all final determinations regarding what is inspected. The system functions exclusively as a decision-support tool, providing informational input and non-binding recommendations to inspectors.

54. Will this project have major impacts on staff, either in terms of their numbers or their roles?
No [Points: +0]

55. Describe the criteria used to evaluate client data and the operations applied to process it.

The ERA-Food model concept is based on the allocation of risks to food establishments based on their impact on consumers' health in Canada. Results are expressed in Disability Adjusted Life Years (DALYs), a unit that is used internationally by government and public health organizations to identify and quantify the impact on a population so they can prioritize prevention and management efforts accordingly. This value takes into consideration the number of illnesses associated with a food safety hazard, their attribution to specific food commodities and sub-products, and the health impact per case of illness for each hazard. Thus, the health impact is first allocated to individual establishments based on the type of activity and the volume of each product type they manufacture. The establishment-level health impact is then adjusted considering the presence or absence of specific food safety risk factors in the establishments and their relative risk.

56. Describe the output produced by the system and any relevant information needed to interpret it in the context of the administrative decision.

The outputs of the ERA model are the inherent, mitigated and final risk results which are displayed as a graph with three bars representing the risk level associated to an establishment (available in the ERA Establishment Risk Profile). The first bar represents the risk level when considering the inherent risk factors such as the type of products and the volume. The second bar represents the risk level when considering the inherent risk factors combined with the mitigation factors such as third-party audits. The last bar represents the final risk considering the inherent risk factors, mitigation factors and compliance factors. Compliance factors include information about how well the establishment has been complying with regulatory requirements. Higher risk establishments or sectors that require urgent attention can be easily identified and focused on using this output. The agency provides interpretation materials to help both regulated parties and inspectors understand and apply the ERA-Food risk-assessment outputs. Industry has access to publicly available guidance material and training explaining how results relate to their establishment, while inspectors receive additional internal guidance and training to ensure consistent application of the model. These resources support transparency for industry, and consistency in regulatory decision making.

57. Will the system perform an assessment or other operation that is not currently completed by a human?
No [Points: +0]

58. Is the system used by a different part of the organization than the ones who developed it?
Yes [Points: +2]

59. Are the impacts resulting from the decision reversible?
Reversible [Points: +1]

60. How long could impacts from the decision last?
Some impacts may last a matter of months, but some lingering impacts may last longer [Points: +2]

61. Describe why the impacts resulting from the decision are as selected, including potential risks identified and mitigations in place.
There are three kinds of risk factors that influence the ERA-Food model result: inherent, mitigation and compliance factors.

Information linked to the inherent and mitigation factors is provided by the regulated parties (i.e., food industry) when they complete an online questionnaire while applying for their licence. This information can be updated by them at any time, for example, when a change of activities occurs on their side.

Information related to the compliance factors is considered variably. These factors could impact the risk result for a period of one, two, three or up to five years depending on the type of information considered (e.g., inspection results, recalls). This information is updated at any time new information becomes available, for example, through the recording of inspection results in the CFIA data systems. It is important to note that inspectors can always adjust the food program suggested inspection scope after an on-site walkthrough and assessment of conditions observed within the establishments.

62. The impacts that the decision could have on the rights or freedoms of individuals may be:
Little to no impact [Points: +1]

63. Describe how this impact level was determined. Include which human rights frameworks were assessed, potential risk that were identified and mitigations put in place.
The ERA-Food model assesses the food safety risk of establishments and has no impact on the rights or freedoms of individuals.

64. The impacts that the decision could have on the equality, dignity, privacy, and autonomy of individuals may be:
Little to no impact [Points: +1]

65. Describe why the impacts resulting from the decision are as selected, including potential risks identified and mitigations in place.
The ERA-Food model assesses the food safety risk of establishments and has no impact on the equality, dignity, privacy, and autonomy of individuals.

66. The impacts that the decision could have on the health and well-being of individuals may be:
Moderate impact [Points: +2]

67. Describe why the impacts resulting from the decision are as selected, including potential risks identified and mitigations in place.

As part of the Canadian Food Inspection Agency's efforts to modernize its inspection system, the ERA-Food model was developed to evaluate food establishments based on the level of food safety risk they represent to consumers. The model uses data and a mathematical algorithm to determine the level of risk to inform oversight required by inspectors. This is part of the Agency's mission to safeguard food to enhance the health and well-being of Canada's people.

68. The impacts that the decision could have on the economic interests of individuals may be:
Little to no impact [Points: +1]

69. Describe why the impacts resulting from the decision are as selected, including potential risks identified and mitigations in place.

The ERA-Food model assesses the food safety risk of establishments. It does not consider economic risks of food establishments and as a result has no impact on the economic interests of individuals.

70. The impacts that the decision could have on the ongoing sustainability of an environmental ecosystem, may be:
Little to no impact [Points: +1]

71. Describe why the impacts resulting from the decision are as selected, including potential risks identified and mitigations in place.

The ERA-Food model assesses the food safety risk of establishments and has no impact on the ongoing sustainability of an environmental ecosystem.

72. Have you assessed system performance for clients with a range of personal and intersectional identity factors (for example, gender, age, race, language, ethnicity, disability) to verify that decisions and outcomes are fair for a diverse group of people, and have the identified disparities been addressed?
Assessment conducted; no performance differences identified. [Points: +0]

73. Describe your approach to assessing system performance.
Assessment of system performance across diverse populations is informed by the nature of the data collected from regulated parties and the scope of the ERA-Food model. The model evaluates risk as the establishment level and does not process personal information related to individual identity factors (e.g. gender, age, race, ethnicity, or disability). As such, direct assessment or performance across these identity factors is non-existent.

However, when completing the Food Safety Supplemental Information (FSSI) questionnaire, licence applicants are asked whether their establishment(s) distribute food to vulnerable population groups (e.g. hospitals, long-term care facilities). This variable was used as a proxy to assess whether the model produces differential outcomes for establishments serving higher-risk populations. Specifically, the proportion of establishments indicating distribution to vulnerable populations was analyzed relative to the total regulated establishment population, and their assigned risk categories were compared to those of establishments within the same sector that do not serve these populations. This comparative analysis did not identify systematic differences in model outputs that would suggest unfair or biased treatment.

These results indicate that the ERA-Food model is performing consistently across the assessed groups. Ongoing monitoring is in place to ensure

potential disparities are identified and addressed, and to support continuous improvement as data availability evolves.

74. Have you tested system performance for clients with a variety of different disabilities?

No

[Points: +1]

75. Describe why system performance has not been assessed across a variety of disabilities.

System performance has not been assessed across a variety of disabilities because the ERA-Food model is used to evaluate food safety risk at the establishment level, not at the individual level.

76. Are there clients or groups of clients that will experience the most significant negative impacts from the system being used?

No

[Points: +0]

77. Describe

No specific clients or groups of clients are expected to experience disproportionate negative impacts from the use of the ERA-Food model. The model supports the CFIA's risk-informed approach to managing food safety risks by informing inspection prioritization and work planning. External clients (i.e. Safe Food for Canadians (SFC) licence holders with domestic activities) are assessed using a standardized methodology and consistent risk criteria. Risk results reflect the level of risk associated with the products and activities conducted at each establishment, rather than characteristics of specific groups, ensuring fair and consistent treatment across regulated parties.

For internal clients (e.g. inspectors, analysts, risk managers), the model provides structured, evidence-based risk insights that support more consistent and transparent decision-making. While this may influence how inspection activities are prioritized, it reduces variability as a consistent process is applied, enhancing alignment with risk-based objectives.

Establishments identified as higher risk may be subject to an increased inspection frequency or oversight, however, this reflects their risk and is consistent with the CFIA's mandate of protecting Canadian consumers.

Ongoing monitoring and validation processes are in place to ensure the model continues to operate as intended and does not introduce unintended or disproportionate impacts.

78. What proportion of clients will the system be applied to?

100%

[Points: +0]

79. The impact of individuals being wrongfully included or overlooked by the system could be:

Moderate impact

[Points: +2]

80. Describe why the impact of being wrongfully included or overlooked by the system is as selected.

Only SFC licence holders with domestic activities are evaluated by the ERA-Food model. If an SFC licence applicant mistakenly adds domestic activities to their licence, they will be asked to complete the FSSI (formerly AEI) questionnaire during the licencing process. As part of the licence application review process, an agency agent will reach out to the licence holder to confirm their activities, and at this point the error will be identified and corrected. Thus,

the chances of being wrongfully included are virtually zero.

There is potential that a SFC licence holder mistakenly omits domestic activities from their licence application (e.g. if previously they only performed import activities but recently started conducting domestic activities). In this case, there is a risk that this establishment will not be evaluated by the ERA-Food model. As a result, the appropriate inspection frequency will not be applied which may lead to regulatory blind spots where critical risks may go unmonitored, and incidents or non-compliance are more likely to go unnoticed.

About the Data

81. Will the system use personal information as input data?

Yes

[Points: +2]

82. Have you verified that the collection of personal information is limited to only what is directly related to a program or activity of your institution?

Yes

[Points: +0]

83. Is the personal information of individuals used in a decision-making process that affects those individuals?

Yes

[Points: +2]

84. Have you verified whether the system is using personal information in a way that is consistent with: (a) the current personal information banks (PIBs) and privacy impact assessments (PIAs) of your programs or (b) planned or implemented modifications to the PIBs or PIAs that take new uses and processes into account?

Yes

[Points: +0]

85. List relevant PIB numbers.

CFIA PPU 103

86. Will the personal information be shared (check all that apply)

Within your department

[Points: +1]

87. Is the training and testing data for the system representative of the clients being served?

Yes

[Points: +0]

88. Describe how you have assessed that data was representative.

A performance assessment of the ERA-Food model was conducted prior to its implementation. The objectives of this exercise were to:

- Evaluate data collection tools
- Estimate the level of agreement between the risk assessment outputs obtained with the ERA-Food model and the assessment done by CFIA senior inspectors/experts for the same establishments
- Evaluate if the differences between the two assessments (if any) are associated with a specific parameter of the model
- Refine the model, if needed, based on the identification of major discrepancies.

To do so, a stratified random selection of establishments within each commodity and across Canada was conducted to ensure national representativeness. When necessary, the number of establishments drawn

from each group was proportionally aligned to their distribution within the commodity. In addition, the sampling design accounted for establishment size, ensuring a balanced representation of small, medium, and large operations to better reflect the diversity of the sector.

Information from the Food Safety Supplemental Information (FSSI, formerly the Additional Establishment Information (AEI)) questionnaire, as well as data from internal CFIA systems – such as Digital Services Delivery Platform (DSDP), Incident Management System (IMS), Compliance Verification System (CVS), and the Transparency Initiative were used and evaluated as data sources to collect data on the establishment's historical performance.

89. Are there biases in the training data that could increase the likelihood of discriminatory outcomes or lead to other impacts on human rights?

No

[Points: +0]

90. Explain the process taken to confirm that there are no biases in the training data that could increase the likelihood of discriminatory outcomes or other impacts on human rights.

The ERA-Food automated decision tool does not rely on machine-learning or any type of model that is trained using historical data. Instead, it uses a set of predefined business rules. When information about an establishment is updated in the system, the tool simply re-applies these rules to the latest data. This means the results are refreshed automatically, but the underlying logic never changes or learns from data.

In addition, to ensure fair and consistent assessment across all food businesses, prior to the model implementation, the ERA-Food model was tested through pilot projects across all food sectors, and the performance of the model assessed through a validation exercise that compared the algorithm's outputs to the assessment done by senior inspectors.

Many food safety experts also participate(d) in the expert elicitation exercises and a Scientific Advisory Committee (SAC) endorsed the methodologies followed. This reduced potential bias in risk identification, attribution, and weighting.

The ERA-Food model only uses and generates data at the business level (e.g., type of product manufactured, volume, risk category of the establishment), and does not include personal or demographic information related to individuals or groups. As such, it does not differentiate external clients based on identity characteristics, and it does not produce outputs that could result in differential treatment of individuals. For internal clients, the model provides standardized, objective risk scores that support consistent and transparent decision-making.

Overall, the combination of rule-based design, expert validation, and the absence of individual data supports the conclusion that there are no identifiable biases in the data or methodology that would increase the likelihood of discriminatory outcomes or adverse impacts on human rights.

91. What is the highest security classification of the input data used by the system? (Select one)

Protected B

[Points: +3]

92. Are there measures in place to ensure the accuracy of the personal information used?

Yes

[Points: +0]

93. If yes, describe the measures taken to ensure and verify the accuracy of personal information.

To ensure the information contained in the ERA-Food profile is accurate and up to date, a high-level review of the licence applicant information, including their establishment's activities and mitigation measures is conducted by staff at CFIA (i.e., the National Center for Permissions) during the application, renewal and amendment of an SFC licence. Additionally, the inspectorate is advised to review the ERA risk profile with licence holders during their first Preventive Control Inspection, and licence holders are also invited to update their FSSI (formerly AEI) as new information becomes available.

94. Who controls the data?

Federal Government

[Points: +1]

95. Will the system use data from multiple different sources?

Yes

[Points: +2]

96. Will the system require input data from a network-connected device? (such as Internet of Things, sensor)

Yes

[Points: +4]

97. Will the system interface with other IT systems?

Yes

[Points: +4]

98. Who collected the data used for training the system?

Your institution

[Points: +1]

99. Who collected the input data used by the system?

Your institution

[Points: +1]

100. Describe the input data collected and used by the system, its source, method of collection, and any quality assurance measures in place.

To collect information on the establishment's inherent and mitigation factors, regulated parties fill out an online questionnaire (FSSI (formerly AEI)) through My CFIA. When applying for a Safe Food for Canadians (SFC) licence, all food businesses that conduct domestic activities must list their operational activities, as well as information about how risk is controlled at their establishment via the FSSI. This data along with the establishment's compliance information extracted from CFIA internal data systems (e.g., inspection results, food safety recalls and complaints, control/enforcement actions) are centralized in the CFIA Enterprise Data Warehouse using an Extract, Transform, Load (ETL) process.

101. Will the system require the analysis of unstructured data to make an assessment, recommendation or decision?

No

[Points: +0]

102. Has there been a decision to exclude certain data?

No

[Points: +0]

Section 3.3: Mitigation Questions and Answers

Consultations

1. Internal Stakeholders (federal institutions, including the federal public service)

Yes [Points: +2]

2. Which internal consultees or partners have you engaged?

Strategic Policy and Planning

Data Governance

Program Policy

Legal Services

Access to Information and Privacy

Communications services

Client Experience / Client Relationship Management

Other (describe)

3. Please describe

Corporate Management Security, International Affairs, Operations, Other Government departments (Public Health Agency of Canada and Health Canada)

4. External consultees or partners (groups in other sectors or jurisdictions).

Yes [Points: +2]

5. Which external consultees or partners have you engaged?

Academia

Industry

Governments in other Jurisdictions

6. Identify the consultees or partners and indicate the project lifecycle stage (Concept, Design, Data collection and processing, Building or adapting the model, Testing, Deployment, Production, Monitoring, Decommission) in which engagement took place.

Academia (Université de Montréal, University of Guelph, Dalhousie university, University of Manitoba) and industry (Risk Sciences International): Consulted at all project lifecycle stages of the ERA-Food Model.

Government in other Jurisdictions including foreign and provincial food safety authorities (e.g., Ontario Ministry of Agriculture, Food and Agribusiness (OMAFRA), Ministère de l'Agriculture, des Pêcheries et de l'Alimentation (MAPAQ), Belgian Food Safety Authority, Agence nationale de sécurité sanitaire) were engaged at different stages of the project lifecycle including the concept and design (through an international benchmarking exercise) and deployment.

7. Did you consult with the clients that will be most impacted by the system?

Yes [Points: +3]

8. Did you consult with clients or communities that will be adversely impacted by the system?

Yes [Points: +3]

9. Have consultees or partners raised any significant concerns about the system in the following areas:

Fairness of outputs

Protection of personal information and/or other privacy concerns
Accuracy

10. Describe how the identified concerns have been assessed and managed.

Concerns regarding fairness of outputs, protection of personal information as well as accuracy of the system were raised during the initial design and implementation of the model. The concerns encountered were primarily related to understanding the model's functioning. A continuous support approach, comprising training, industry question and answer sessions, as well as individual consultations, were employed to address them. Over time, stakeholders have become increasingly comfortable with the model, and no significant concerns have been raised since then.

11. Explain how you documented, analyzed and addressed the feedback received.

Feedback from stakeholders and partners is documented through inquiries submitted to the ERA technical team at cfia.eramodel-modeleere.acia@inspection.gc.ca. The CFIA reviews and analyzes the feedback, and addresses identified concerns through updates to the model and guidance on an ongoing basis, using the most up-to-date science, technology, and risk assessment approaches.

De-risking and Mitigation Measures - Data Quality

12. Do you have documented processes in place to test datasets against biases, discrimination, and other unexpected outcomes? This could include experience in applying frameworks, methods, guidelines, or other assessment tools.

Yes [Points: +3]

13. Is the documented process publicly available?

No [Points: +0]

14. Have you developed a process to document how data quality issues were resolved during the design process?

Yes [Points: +1]

15. Is the process publicly available?

No [Points: +0]

16. Have you undertaken a Gender-based Analysis Plus (GBA Plus) of the data?

Yes [Points: +2]

17. Are the GBA Plus findings publicly available?

No [Points: +0]

18. Summarize the findings or provide a link if publicly available.

The GBA Plus analysis determined that the ERA model does not need Gender-Based considerations, as the information collected, and the results produced focus exclusively on compliance requirements and operational standards of the Food Operations Division. The model's scope is centered on federally regulated food establishments and does not directly or indirectly interact with or affect communities, groups, or individuals in ways that would require specific Gender-Based considerations.

19. Do you have a documented process to manage the risk that low quality, outdated or unreliable data is used to make an automated decision?

Yes [Points: +2]

20. Is the process publicly available?

No [Points: +0]

21. Will the data used for this system be posted on the Open Government Portal?

No [Points: +0]

De-Risking and Mitigation Measures - Procedural Fairness

22. Does the audit trail provide information to trace the chain of legislative authority for the decision and key decision points?

Yes [Points: +1]

23. Are all key decision points in the automated system's logic linked to the relevant legislation, policy or procedures?

Yes [Points: +1]

24. Does the audit trail capture all the recommendations or decisions made by the system?

Yes [Points: +2]

25. Are all key decision points identifiable in the audit trail?

Yes [Points: +2]

26. Is there a process in place to log the instances when overrides were performed?

No [Points: +0]

27. Is there an up-to-date log of the changes made to the model and the system?

Yes [Points: +2]

28. Does the audit trail capture information that can be used to help generate an explanation of the decision where required?

Yes [Points: +1]

29. Is the system able to produce reasons for its decisions or recommendations when required?

Yes [Points: +2]

30. Does the audit trail identify precisely which version of the system was used for each decision it supports?

No [Points: +0]

31. Does the audit trail show who the authorized decision-maker is?

No [Points: +0]

32. Is there a mechanism to capture feedback by users of the system?

Yes [Points: +2]

33. Describe how user feedback informs updates to the system.

From the perspective of industry users, the FSSI (formerly AEI) includes an

“Others” option that allows establishments to report additional processes or mitigations steps not covered by standard categories of the ERA-Food model. These entries are reviewed by the ERA team to determine whether updates or refinements to the FSSI are needed.

Feedback is also gathered through inquiries or comments sent to the generic inbox of the ERA-Food technical team. These messages can trigger improvements to the FSSI– for example, when users report IT problems accessing the FSSI that are then addressed in the system by the ERA-Food technical team.

34. Is there a recourse process established for clients to challenge the decision?

Yes [Points: +2]

35. Provide information on the recourse process or link to a website where this is described.

When a client challenges their ERA results, they are asked to contact the ERA-technical team. A meeting is then arranged to review their risk profile step-by-step, verify the accuracy of the information and provide a detailed explanation on how their ERA outcome was determined by the model.

36. Does the audit trail include change control processes to record modifications to the system's operation or performance?

No [Points: +0]

37. Does the system base its decision or recommendation on all of the information provided by the client?

No [Points: +0]

38. Have you undertaken mitigations in algorithm development and model testing to address bias and discrimination?

Yes [Points: +2]

39. Describe the metrics used and any mitigations taken to address identified issues.

Prior to the model implementation, the ERA-Food model algorithm was tested through pilot projects across all food sectors, and the performance of the model assessed through a validation exercise that compared the algorithm's outputs to the assessment done by senior inspectors.

40. Does the system only consider input data and information relevant to the decision?

Yes [Points: +2]

41. Are the system rules directly relevant to the administrative decision?

Yes [Points: +2]

42. Indicate additional procedural fairness protections in place (check all that apply):

Clients are provided with adequate information about the decision-making process and have the opportunity to meaningfully participate [Points: +1]

Clients are provided with all documents and information in which the administrative decision was based [Points: +1]

Mechanism for assessing and responding to client feedback [Points: +1]

43. Are there processes in place to ensure impartiality and ethical conduct by the decision-maker in the use of the system and its outputs?

Yes [Points: +2]

44. Have you undertaken training on appropriate use and awareness of risks associated with automation bias?

Yes [Points: +2]

45. Will your team be regularly retraining the model or reviewing the rules?

Yes [Points: +2]

46. Do you have monitoring processes in place to detect changes in system performance over time?

Yes [Points: +2]

47. Do you have a process in place to update the system based on the monitoring results?

Yes [Points: +1]

48. Describe

No model retraining is applied to the ERA Food model. Adjustments/updates to the ERA-Food model algorithm rules are applied continuously and the process to update the system depends on the needs identified (e.g., new information on a pathogen, a new mitigation strategy applied by the industry). If there are many adjustments/updates, they will be usually grouped and applied, for example, at the beginning of the new fiscal year. In this case, as part of the governance structure, the ERA technical team works with a Scientific Advisory Committee (SAC) which is consulted on a yearly basis to refine, adjust and adapt the model ensuring the scientific rigour is respected. The team follows a Standard Operating Procedure for the implementation of any adjustment to the algorithm. When they are implemented, some changes in the risk result/category of a particular establishment may occur, compared to the previous calculation for the same establishments (i.e. before the algorithm update). The ERA technical team communicates with inspectors and industry whenever such updates are implemented, by using the communication processes already established (e.g., through listservs or info sessions).

In contrast with ERA-Food model algorithm updates, the ERA establishment risk results are updated daily through the ERA business solution that was launched in April 2021. If an establishment updates its FSSI (formerly AEI), inspectors will be able to see the updated risk result for this establishment the next day. The same applies for new inspection results reported for an establishment: upon entering them into DSDP, the ERA business solution will extract them overnight and will update the risk result accordingly. Continuous monitoring of the model outputs follows the Standard Operating Procedure developed for this purpose.

De-Risking and Mitigation Measures - Privacy

49. If your system uses or creates personal information, have you undertaken a PIA, or updated an existing one?

Yes [Points: +2]

50. Indicate the following in your answer: Title and scope of the PIA; How the automation project fits into the program; and Date of PIA completion or modification.

ERA System Project

Privacy Impact Assessment Questionnaire

Date of Privacy Impact Assessment completion: March 6, 2017

The initiative supports the existing Program entitled "Food Safety Program" under the Strategic Outcome "A safe and accessible food supply and plant and animal resource base". ERA is a service that supports this program.

51. Have you designed and built security into your systems from the concept stage of the project?

Yes

[Points: +2]

52. Describe

As part of the project development, the ERA-Food automated tool was subjected to a Privacy Impact Assessment (PIA) in March 2017, and it was determined that information was used in a way that is consistent with the current Personal Information Banks, Privacy Practices and Security Management.

The ERA and Digital Service Branch (DSB) teams designed and built security conditions into the system from the concept stage of the project. ERA data is controlled by the federal government (CFIA) and actions have been implemented for the use and transfer of the model inputs and outputs (e.g., Protected B risk profiles).

53. Have you designed and built privacy into your systems from the concept stage of the project?

Yes

[Points: +2]

54. Describe

As part of the project development, the ERA-Food automated tool was subjected to a Privacy Impact Assessment (PIA) in March 2017, and it was determined that information was used in a way that is consistent with the current Personal Information Banks, Privacy Practices and Security Management.

The ERA and DSB teams designed and built privacy conditions into the system from the concept stage of the project. ERA data is controlled by the federal government (CFIA) and actions have been implemented for the use and transfer of the model inputs and outputs (e.g., Protected B risk profiles). Data stewardship principles do not allow the FSSI (formerly AEI) to be used for any purpose other than for the purpose it was communicated for. The ERA National Manager is the data steward for ERA and FSSI data. The ERAM team has a formal agreement in place with the DSDP and IMS data stewards for the appropriate use of DSDP and IMS data respectively. The FSSI preamble available to clients clearly state how the collected data is to be used, accessed and stored.

55. Is the information used within a closed system (no connections to the Internet, Intranet or any other system)?

No

[Points: +0]

56. If the disclosing of personal information is involved, has an agreement or arrangement with appropriate safeguards been established?

Yes

[Points: +1]

57. Will you de-identify any personal information used or created by the system at any point in the lifecycle?

No

[Points: +0]